



California State University Chico

Hierarchical Web of Hypernetwork Experts for mitigating catastrophic forgetting in Audio Classification

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Problem Statement

While monolithic deep neural networks offer significant representational power, they are frequently susceptible to catastrophic forgetting when introduced to new tasks, as the optimization process often overwrites previously learned feature maps.

I propose a Hierarchical Mixture of Experts architecture structured as a Directed Acyclic Graph. In this framework, experts function as nodes that transmit information to their children. Utilizing a Neat-Inspired Evolutionary Algorithm, the system dynamically spawns child experts that utilize hypernetworks to interpret and build upon the feature maps of their parent nodes.



Architectures

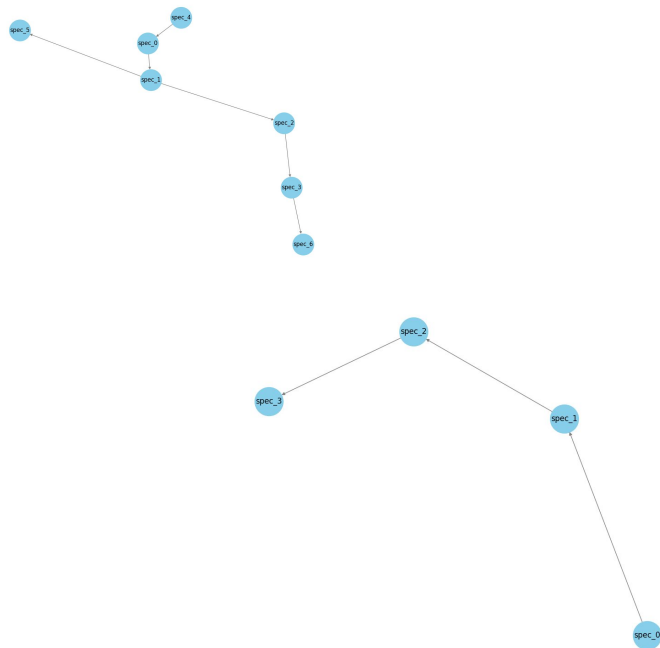
Each node within the graph consists of a Residual Squeeze and Excitation Block.

The residual component employs a sequence of 1D convolution, batch normalization, ReLU activation, and pooling. To facilitate information transfer, the layer dimensions are dynamically mapped, `in_channels` are derived from the parents expert's output, while `out_channels` define the current expert's feature space.

This is augmented by a Squeeze-Excitation block compromising dual fully connected layers with ReLU and Sigmoid activations, enabling the model to perform channel-wise feature recalibration on the inherited parent maps.



Evolution Architecture



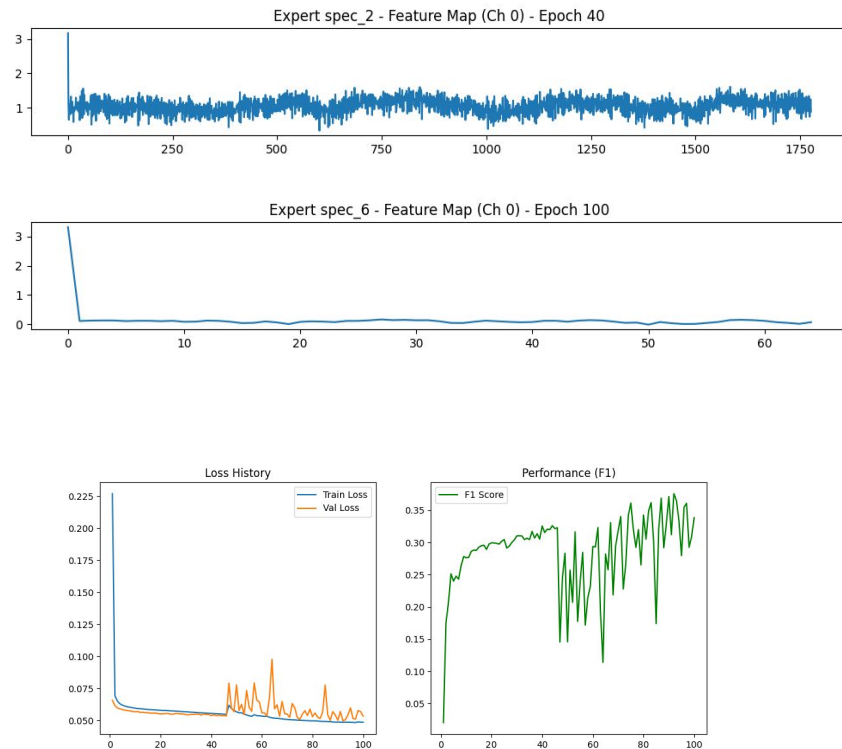
The NEAT inspired algorithm is enacted after a set stagnation threshold, if the model does not improve then a mutation kicks in. There are four mutations, add a new expert at the root of the hierarchy, spawn a node from an existing expert, remove a child to optimize and remove underperforming experts, and add connection that establishes a new link between existing nodes.



Results and Findings

While the proposed architecture did not surpass the 43% F1 score set by the baseline ResSENet, it demonstrated a capacity for growth in response to performance plateaus. On the MagnaTagATune data, comprising of 188 labels, the model achieved a respectable 37% F1 Score.

A notable observation was the performance dip following a mutation. The introduction of a new expert initially disrupted the existing feature landscape, causing a temporary decline in accuracy. The model was able to stabilize in subsequent epochs.



Summary and Conclusions

The “Hierarchical Web of Hyper-Experts” serves as a successful proof of concept for an evolutionary network capable of acquire new task knowledge while mitigating the effects if catastrophic forgetting through a modular graph structure.

Future advancements should work on fine-tuning the cooldown period and stagnation thresholds to ensure the graph has sufficient time to integrate and stabilize new nodes before mutations occur. Current experts initialize with random noise and specialize over time, future work will explore pre-trained specialized experts to provide a more robust starting point.



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Thank You
